

Supplemental Methods: STATE

1 STATE

We benchmarked the State Embedding (“SE”) stage of STATE[1], a self-supervised cell-embedding transformer trained with a tabular energy loss over masked gene-expression “sentences”. Each cell is tokenized into a sequence of (gene, count) pairs, gene tokens are mapped to fixed protein-language-model embeddings, and the sequence is processed by a transformer encoder; a reconstruction decoder predicts per-gene expression values across the masked positions.

Following the published recipe, gene tokens are replaced with pretrained protein embeddings from TranscriptFormer[2], a cross-species single-cell foundation model that encodes each gene using ESM-2[3] protein language model embeddings. We use the 2,560-dimensional gene embeddings released by TranscriptFormer for both *Mus musculus* and *Homo sapiens*, merged into a single multi-species table.

To make STATE comparable to the Geneformer baseline used elsewhere in this study, we override the published SE defaults to a smaller architecture (Table 1). The full set of hyperparameters used for from-scratch training is reported in Table 1.

Unlike Geneformer and SCVI, STATE training duration is governed by a fixed *step* budget chosen so that the total amount of optimization remains roughly constant across the sweep, with step-based early stopping acting as an upper bound. After training, the best checkpoint is selected by minimum validation loss, and cell embeddings are extracted from this checkpoint for downstream mutual-information estimation.

Figure 1 shows representative training and validation loss curves for SE trained on PBMC at the largest dataset size (46,415 cells) and full quality. Both losses decrease monotonically and the validation curve closely tracks training, indicating stable convergence without overfitting.

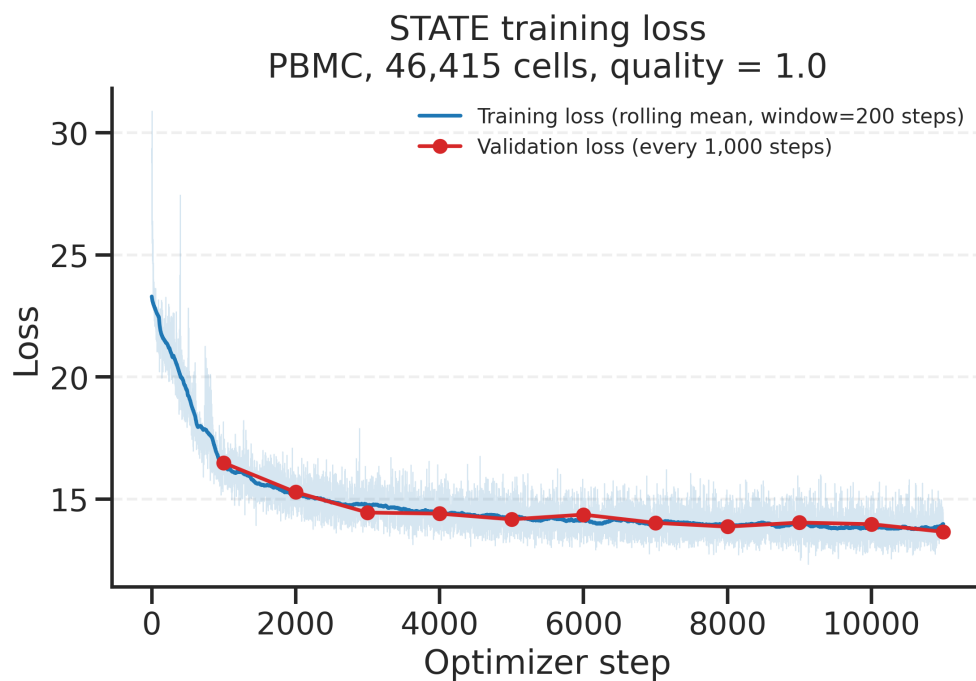


Figure 1: Training and validation loss curves for STATE SE trained on PBMC (46,415 cells, quality = 1.0), shown on linear axes. Validation is computed every 1,000 optimizer steps. The raw per-step training loss is plotted faintly in the background, and the bold blue curve overlays a centered rolling mean (window = 200 steps) to suppress minibatch noise; no other smoothing is applied. The validation curve is shown unsmoothed (one point per evaluation).

Table 1: Hyperparameters for STATE (SE) as used in this study.

Parameter	Value
<i>Model architecture</i>	
Number of transformer layers	3
Hidden dimension	256
Attention heads	4
Feed-forward dimension	512
Cell-embedding (output) dimension	256
Maximum sequence length	2,048 tokens
Gene token embedding	TranscriptFormer protein embeddings (ESM-2, 2,560-dim, mouse + human)
Token-level objective	20% variable masking
Reconstruction head	Per-gene decoder + count head
Loss	Tabular energy loss (energy distance, gene- and cell-level)
<i>Optimization</i>	
Optimizer	AdamW
Maximum learning rate	1×10^{-4}
Weight decay	0.01
Dropout	0.1
Maximum gradient norm	0.8
Global batch size	64
LR schedule	Linear warmup (0.33 $\rightarrow$ 1.0, 3% of steps) + cosine decay
<i>Training and checkpointing</i>	
Validation interval	1,000 optimizer steps
Validation batches per check	50
Best-checkpoint selection	Minimum validation loss
Early stopping	Step-based, monitor: validation loss

2 Fine-Tuning the Pretrained SE-100M Model

In addition to training STATE from scratch, we evaluated fine-tuning the publicly released Arc Institute SE-100M embedding model[1] on the PBMC noise-scaling grid. SE-100M was pretrained on a large multi-dataset corpus using the full STATE self-supervised objective with dataset-level batch correction. To fine-tune SE-100M, we first downloaded the pretrained model checkpoint and stripped all optimizer and scheduler state, resetting the epoch and step counters to zero while retaining the model weights. This ensures that fine-tuning begins from the pretrained representation with a freshly initialized optimizer and learning-rate schedule. For each (size, quality) cell in the grid, gene token mappings were re-generated using SE-100M’s own protein embedding vocabulary—also based on ESM-2[3]—so that the tokenization is consistent with the pretrained weights. The stripped checkpoint was then placed as the training start point for each cell. Each cell was fine-tuned independently for a fixed 3-epoch budget with early stopping disabled, so all cells received the same number of epochs. The batch-correction head was kept enabled during fine-tuning (matching the pretrained architecture exactly) to avoid weight-shape mismatches at checkpoint load; the head receives a placeholder dataset label signal on PBMC that does not affect the encoder gradients. After fine-tuning, cell embeddings were extracted from the end-of-training checkpoint and mutual information was estimated against `celltype.l3` and `protein.counts`. The fine-tuning hyperparameters are summarized in Table 2.

Table 2: Hyperparameters for fine-tuning SE-100M on the PBMC noise-scaling grid. N denotes the number of training cells.

Parameter	Value
Training epochs	3 (fixed; no scaling with N)
Early stopping	Disabled
Base maximum learning rate	1×10^{-6}
LR scaling (small N)	$\max\left(0.3, \sqrt{N/10,000}\right) \times 10^{-6}$ for $N < 10,000$
LR schedule	Cosine decay (no warmup ramp)
Batch size	64
Gradient accumulation steps	16
Effective batch size	1,024
Dropout	0.15 ($N \geq 10,000$); 0.20 ($N < 10,000$)
Weight decay	0.01 ($N \geq 10,000$); 0.05 ($N < 10,000$)
Checkpoint selection	End-of-training weights (no validation windows in 3-epoch budget)

References

- [1] Adduri, A. K., Gautam, D., Bevilacqua, B., Imran, A., Shah, R., Naghipourfar, M., Teyssier, N., Ilango, R., Nagaraj, S., Dong, M., Ricci-Tam, C., Carpenter, C., Subramanyam, V., Winters, A., Tirukkovular, S., Sullivan, J., Plosky, B. S., Eraslan, B., Youngblut, N. D., Leskovec, J., Gilbert, L. A., Konermann, S., Hsu, P. D., Dobin, A., Burke, D. P., Goodarzi, H., & Roohani, Y. H. (2025). Predicting cellular responses to perturbation across diverse contexts with State. *bioRxiv*. <https://www.biorxiv.org/content/10.1101/2025.06.26.661135v2>
- [2] Pearce, J. D., Simmonds, S. E., Mahmoudabadi, G., Krishnan, L., Palla, G., Istrate, A.-M., Tarashansky, A., Nelson, B., Valenzuela, O., Li, D., Quake, S. R., & Karaletsos, T. (2025). A cross-species generative cell atlas across 1.5 billion years of evolution: the TranscriptFormer single-cell model. *bioRxiv*. <https://www.biorxiv.org/content/10.1101/2025.04.25.650731v2>
- [3] Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., & Rives, A. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130.
- [4] Theodoris, C. V., Xiao, L., Chopra, A., Chaffin, M. D., Al Sayed, Z. R., Hill, M. C., Mantineo, H., Brydon, E., Zeng, Z., Liu, X. S., & Ellinor, P. T. (2023). Transfer learning enables predictions in network biology. *Nature*, 618(7965), 616–624.